

# Noor Fatima

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## EDUCATION

### University of Engineering and Technology, Lahore, Pakistan

2023 – Expected 2027

*BS in Computer Engineering | CGPA: 3.83/4.0 (Top 3 in session)*

Research Supervisor: Prof. Dr. M. Shahbaz

## EXPERIENCE

### Research Assistant

Aug 2026 – Present

*University of Saskatchewan – KAIT Collaborative Research Project*

- Diagnosed systematic connector–wire confusion in a **SegFormer-B5** semantic segmentation model for deformable linear objects and designed targeted training interventions for three-class RGB segmentation.
- Investigated boundary errors, small-object recognition, and generalization across held-out wire assemblies, isolating the source of failure without modifying the training dataset or model architecture.
- Contributed to a multi-university research collaboration between the **University of Saskatchewan** and other Canadian institutions under the **Korea Institute for Advancement of Technology (KAIT)**.

### Research Intern

Jun 2026 – Sep 2026

*MITACS Globalink Research Internship, University of Saskatchewan, Saskatoon, Canada*

- Will research a **hybrid AI–Quantum framework** for efficient variational circuit design and optimize ansatz structure using PennyLane/Qiskit.
- Will benchmark hybrid approaches against standard VQE/QAOA baselines and collaborate with an international team to produce **publication-ready results**.

### Research Intern

May 2025 – July 2025

*National Center for Quantum Computing (NCQC), Lahore, Pakistan*

- Developed **QAOA-based optimization algorithms** for NP-hard feature selection problems in healthcare diagnostics, processing 2000+ patient samples.
- Implemented Variational Quantum Classifier (VQC) achieving **98.4% accuracy**, demonstrating **quantum advantage** over classical baselines (+3.7% improvement).

### Research Assistant

June 2024 – Present

*Al-Khawarizmi Institute of Computer Science (KICS), Lahore, Pakistan*

- Built end-to-end **ML pipelines** for EEG-based emotion recognition across 3 datasets (150+ subjects).
- Developed predictive models for Alzheimer's/MCI classification using 80+ patient EEGs and integrated computational neuroscience models for improved stress-decoding.

### Machine Learning Intern

Mar 2024 – Jun 2024

*Datalabb, Lahore, Pakistan*

- Designed and optimized U-Net segmentation framework for 10k+ medical images, achieving **90% mean Dice score**.
- Fine-tuned domain-adapted LLMs for clinical text, improving reliability by **22%** and delivering production-ready models.

### Machine Learning Fellow

Jun 2024 – Sep 2024

*Bytewise Limited, Lahore*

- Selected among top **9% of 2,300+ applicants**; completed 8+ ML modeling and optimization projects.
- Automated preprocessing workflows, reducing runtime from 2 days to under 6 hours.

### Research Assistant

Jul 2025 – Present

*Dpoint Technologies Ltd, Cyprus*

- Researched Predictive Maintenance systems for industrial cranes, improving operational reliability and reducing downtime.
- Analyzed sensor data (vibration, temperature, current) to identify patterns leading to mechanical failures.
- Developed ML models for anomaly detection and fault prediction, integrating results into real-time dashboards.

**Teaching Assistant**

Spring 2026

*University of Cyprus*

- Assisted in teaching **Programming Fundamentals** and **Information Security**.
- Supported students through lab sessions, assignments, and conceptual clarifications.

**Teaching Assistant**

Fall 2025

*University of Cyprus*

- Assisted in delivering **Software Engineering** and **Computer Graphics** courses.
- Guided students in project development, debugging, and applied problem-solving.

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SELECTED PROJECTS

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**NeuroAI Platform | Deployed App | GitHub**

June 2025 – Present

*Tech Stack: Python, PyTorch, MNE, ReactJS, Docker, AWS*

- Built scalable ML platform with automated parameter prediction for EEG model benchmarking, reducing testing time from **2–3 days to minutes**.
- Implemented **intelligent system** for data ingestion and annotation workflows, tested by 5+ researchers in production environment.

**Complete Blood Cell Counter | Deployed App | GitHub**

2025

*Tech Stack: Python, PyTorch, YOLOv11, OpenCV, ONNX, Gradio, Docker*

- Curated and merged **1,294** annotated microscope images from two public datasets into a unified YOLO-format pipeline with automated integrity checks for corrupt labels, invalid classes, and image-label mismatches.
- Trained a **YOLOv11s** object detector to identify platelets, RBCs, and WBCs in blood smear images, achieving **97.6% mAP@50** and **94.5% precision** on the validation set.
- Exported the model to **ONNX** and deployed an interactive **Gradio** web app for real-time cell detection and counting, containerized with Docker on Hugging Face Spaces.

**Real-time Seizure Detection & Classification**

May 2025 – July 2025

*Tech Stack: Python, PyTorch Geometric, GNNs, Optimization Algorithms*

- Developed GNN-based optimization system for seizure detection on **TUH EEG** dataset, achieving **90% detection**, **87% classification**, and **84% early prediction**.
- Applied advanced **hyperparameter optimization** techniques to approach published clinical benchmarks on complex medical data.

**Energy-Efficient Sleep Stage Classification**

Feb 2025 – March 2025

*Tech Stack: Python, TensorFlow, snnTorch, MNE*

- Implemented Spiking Neural Networks (SNNs) for **ISRUC-Sleep** dataset (118 subjects), optimizing for low-power edge computing.
- Achieved **82% accuracy** while reducing energy consumption by **35%** through systematic parameter optimization for mobile devices.

**Multiview Neural Decoding with Attention**

Nov 2024 – Jan 2025

*Tech Stack: Python, PyTorch, Attention Mechanisms, Parameter Tuning*

- Built attention-based decoder combining temporal, spectral, and spatial features with optimized architecture.
- Outperformed prior work by +2.3% achieving **93.3% accuracy** on Tsinghua & PhysioNet datasets (500+ subjects) through systematic optimization.

**InsightPilot AI | GitHub**

Aug 2025 – Present

*Tech Stack: FastAPI, Next.js, React, TypeScript, Groq, Tavily, Docker*

- Developed a **multi-agent AI research platform** for strategic market analysis and company audits, enabling automated competitor discovery, ICP extraction, and GTM strategy generation.
- Built real-time orchestration pipeline with **SSE streaming**, live web intelligence, and executive report synthesis, supporting scalable founder and consultant workflows.

## Audio-Analyzer: Intelligent Audio Processing Web Application

August 2024

*Tech Stack: Flask, React.js, JWT, MySQL, Groq API, Librosa, Pydub*

- Developed a secure full-stack app for audio diarization, transcription, feature extraction, and chatbot querying via Groq LLM.
- Implemented JWT-based authentication in Flask backend with a responsive React.js frontend.
- Integrated Librosa and Pydub for signal processing and audio feature extraction.

## EduPath: AI-Powered Academic and Relocation Advisor

March 2025

*Tech Stack: Python, NLP, LLMs, Flask, React*

- Built an AI-powered assistant with voice/chat interface using LLMs to recommend academic institutes and cities.
- Applied multi-criteria decision algorithms considering tuition, rankings, lifestyle, and personal preferences.
- Used Flask for API logic and React for user interface design and interaction.

## AI-powered Dental DICOM Annotation System

June 2025

*Tech Stack: Python, PyTorch, U-Net, pydicom, OpenCV, 3D Slicer, Dragonfly*

- Developed a U-Net-based AI model to segment dental CT scans, identifying nerves, roots, and enamel structures.
- Preprocessed 3D volumetric data using pydicom and OpenCV for compatibility and augmentation.
- Visualized segmentation results using 3D Slicer and Dragonfly tools.

## GroqVision: Vision-LLM Powered OCR and Document Analysis Tool

July 2025

*Tech Stack: Python, Streamlit, Groq Chat API, Pillow, Custom CSS*

- Built a Streamlit web app enabling OCR and document understanding from uploaded image files (receipts, certificates, notes).
- Integrated vision-enabled Groq LLM to answer questions and infer document structure or intent.
- Auto-structured OCR output into Markdown format with support for headings, lists, and code blocks.
- Engineered a clean UI with animated progress bars, alerts, full-width previews, and downloadable output.
- Demonstrated versatility in digitizing academic papers, official records, and handwritten documents.

## LahoreLegacy: AI-Powered Historical Guide

May 2025

*Tech Stack: Python, Streamlit, HuggingFace Transformers, OpenAI, FAISS, LangChain, PyMuPDF*

- Designed and developed an AI-powered web app to explore Lahore's history using LLMs and semantic search.
- Implemented FAISS vector search to retrieve relevant information from Wikipedia, websites, and PDF documents.
- Added PDF upload functionality to search and summarize custom documents using LangChain and PyMuPDF.

## LawyerAssistant: Intelligent Legal Support System | GitHub

Feb 2024

*Tech Stack: ReactJS, Flask, Langchain, LLMs, VectorDB, REST API*

- Engineered a full-stack AI legal assistant for lawyers to perform case research, document summarization, and case law referencing.
- Connected a responsive ReactJS frontend with Flask backend via REST APIs, leveraging Langchain for LLM-powered workflows.
- Integrated VectorDB for contextual retrieval and semantic querying of large legal document corpora.

## Madina Visiting Assistant Chatbot

Jan 2024

*Tech Stack: Twilio API, WhatsApp Business, Flask, Python, LLMs, Webhooks, Ngrok*

- Developed an intelligent WhatsApp chatbot to guide Madina visitors on key topics such as religious sites, prayer timings, and local etiquette.
- Integrated Twilio API with a Flask backend to enable multilingual real-time messaging on WhatsApp Business.
- Used webhook-based architecture with Ngrok for live testing and seamless message delivery between users and LLM engine.

## CraneGPT: Industrial AI Chatbot

April 2023

*Tech Stack: LLaMA-2, Langchain, VectorDB, ReactJS*

- Built an AI-powered chatbot to assist industrial crane operators by providing answers from operation manuals and troubleshooting guides.
- Leveraged Langchain with VectorDB to enable context-aware retrieval and multi-language query handling.
- Integrated a ReactJS frontend with LLM backend workflows, achieving 75% faster task resolution during pilot testing.

## AI-Assisted Clinical Exam Preparation

July 2024

*Tech Stack: Python, Meditron LLM, LoRA, QLoRA, PEFT, Transformers*

- Fine-tuned the Meditron large language model using LoRA and QLoRA techniques to replicate expert-level USMLE-style clinical reasoning.
- Automated the generation of multi-step MCQ justifications, significantly reducing manual annotation workload by 40%.
- Validated generated reasoning with domain experts, achieving over 92% accuracy and improving overall learning efficiency.

## NeuroGenesis: Synthetic Brain Tumor Scan Generation

March 2022

*Tech Stack: PyTorch, DCGAN, NumPy, Matplotlib*

- Designed and trained a Deep Convolutional GAN (DCGAN) to generate synthetic MRI images for brain tumor analysis and augmentation.
- Optimized the adversarial training loop, fine-tuning generator-discriminator balance for enhanced visual fidelity and data variety.
- Produced highly realistic scans with >90% authenticity ratings from experienced radiologists, aiding model generalization.

## Breast Cancer Detection System

Aug 2021

*Tech Stack: Python, ANN, Keras, Pandas, Scikit-learn, Seaborn*

- Developed an artificial neural network to classify malignant and benign breast tumors using structured tabular features.
- Implemented k-fold cross-validation and data augmentation strategies to boost robustness and mitigate overfitting.
- Achieved 99% classification accuracy on the UCI Breast Cancer dataset, outperforming traditional baselines.

## Multimodal RAG for Chest X-ray Analysis (MIMIC-CXR) | GitHub

2025

*Tech Stack: Python, CLIP, Groq LLM, FAISS, Streamlit*

- Built a multimodal Retrieval-Augmented Generation system for chest X-ray interpretation using image-text embeddings.
- Implemented CLIP-based cross-modal retrieval over MIMIC-CXR with FAISS for fast similarity search.
- Integrated Groq LLM to generate structured radiology findings and impressions from retrieved clinical cases.

## Emotion Estimation through Brain Signals

Aug 2025

*Tech Stack: Python, MNE, EDFlib, Scikit-learn, Streamlit*

- Built an EEG-based emotion recognition pipeline with preprocessing, artifact removal, and frequency-band feature extraction.
- Trained machine learning models to classify emotional states from EEG signals with interpretable outputs and evaluation reports.
- Developed an interactive Streamlit dashboard for visualization, confusion matrices, and feature importance analysis.

## Variational Quantum Classifier (Quantum ML App)

Jul 2025

*Tech Stack: PennyLane, PyTorch, Streamlit, Scikit-learn, Plotly*

- Designed a variational quantum circuit with entangling gates for binary classification using expectation measurements.
- Integrated hybrid quantum-classical training pipeline with configurable hyperparameters for circuit experimentation.
- Built interactive visualizations for loss curves, decision boundaries, and model performance analysis.

## Distributed Quantum Algorithms Simulator

Jun 2025

*Tech Stack: Python, Qiskit, Streamlit*

- Implemented distributed execution of GHZ, QFT, and Grover's algorithms using circuit partitioning techniques.
- Integrated Qiskit Aer noise models to evaluate scalability, fidelity, and performance under realistic constraints.
- Developed an interactive interface for visualizing circuit decomposition and benchmarking quantum performance.

## Quantum Gate Visualizer

May 2025

*Tech Stack: Python, Qiskit, Streamlit, Plotly*

- Built an interactive Bloch sphere visualization tool to demonstrate quantum gate operations and qubit transformations.
- Enabled real-time simulation of superposition, entanglement, and phase shifts for educational exploration.
- Designed a Streamlit-based interface with 3D Plotly visualizations for intuitive quantum learning.

## End-to-End Quantum Computing & Quantum Machine Learning Platform

2025

*Tech Stack: PennyLane, Qiskit, PyTorch, Streamlit, Plotly, Python*

- Developed an end-to-end **quantum computing platform** integrating variational quantum machine learning, circuit simulation, and educational visualization tools in a unified system.
- Implemented a **Variational Quantum Classifier** with hybrid quantum-classical training, entangling circuits, and expectation-value-based decision modeling.
- Built a **distributed quantum simulation module** supporting GHZ, QFT, and Grover's algorithms with circuit partitioning and noise-aware benchmarking.
- Designed an interactive **quantum visualization engine** for Bloch sphere dynamics, enabling intuitive exploration of superposition, entanglement, and gate operations.
- Integrated all modules into a single **Streamlit-based research environment** with real-time analytics, performance comparison, and experiment configurability.

## GNN-PdM: Graph Neural Network-Based Predictive Maintenance System | GitHub

2025

*Tech Stack: Python, PyTorch, Graph Neural Networks, Pandas, NumPy*

- Developed a graph neural network-based predictive maintenance system for industrial crane systems using sensor data and digital twin modeling.
- Modeled inter-component dependencies as graphs to capture structural and temporal degradation patterns.
- Implemented training pipelines, evaluation metrics, and cross-validation strategies for robust failure prediction.

## NeuroGraph-TSC: Neuro-Inspired EEG Temporal-Spatial Classifier | GitHub

2025

*Tech Stack: Python, PyTorch, GAT, LSTM, EEG Signal Processing*

- Designed a neuro-inspired graph-based deep learning model combining Graph Attention Networks and LSTMs for EEG-based cognitive state prediction.
- Modeled spatial dependencies among EEG electrodes using graph structures for improved feature representation.
- Incorporated temporal sequence learning to capture dynamic brain activity patterns over time.

## OBE Rubrics Management System | GitHub

2024

*Tech Stack: Python, Streamlit, MySQL, Pandas*

- Developed a Streamlit-based academic management system for Outcome-Based Education (OBE) workflows.
- Implemented modules for managing students, CLOs, rubrics, assessments, and performance evaluation.
- Designed MySQL-backed architecture for structured storage and real-time academic reporting.

## Olympics Data Analysis Dashboard | GitHub

2024

*Tech Stack: Python, Pandas, Streamlit, Matplotlib, Data Visualization*

- Built an interactive data analytics dashboard to explore historical Olympic data and trends.
- Performed medal tally analysis, country-wise performance evaluation, and athlete statistics visualization.
- Designed intuitive UI for dynamic filtering and exploration of large-scale sports datasets.

Tech Stack: Python, Streamlit, LangChain, FAISS, LLMs, PyMuPDF

- Developed an AI-powered document question-answering system for research papers and PDFs.
- Implemented FAISS-based semantic retrieval for efficient context extraction from large documents.
- Integrated LLM-based response generation for accurate, context-aware answers to user queries.

Tech Stack: Python, Machine Learning, Geospatial Data Analysis, Hydrological Modeling, Digital Twins, GIS

- Developed a Digital Twin-based flood prediction and risk management system tailored for Pakistan’s climate resilience challenges.
- Integrated AI models, geospatial datasets, and hydrological simulations to forecast flood risks and identify vulnerable regions in real time.
- Transformed complex environmental and climate data into actionable insights for policymakers, NGOs, and disaster management authorities.
- Validated predictive models using open-source climate and hydrological datasets during the prototype development phase.

TECHNICAL SKILLS

**Languages:** Python, C++, MATLAB, LaTeX | **ML/AI:** PyTorch, TensorFlow, Scikit-learn, Optimization Algorithms | **Specialized:** Quantum Computing (QAOA, VQC), Parameter Prediction, Continuous Learning | **Tools:** NumPy, Pandas, Matplotlib, OpenCV, MNE | **Interests:** ML for Quantum Computing, NP-Hard Optimization, Predictive Modeling

PUBLICATIONS

1. **N. Fatima** and G. Nabi, “TriNet-MTL: A Multi-Branch Deep Learning Framework for Biometric Identification and Cognitive State Inference from Auditory-Evoked EEG,” *eNeuro*, 2026. (Accepted, DOI: 10.1523/ENEURO.0265-25.2025)
2. **N. Fatima** and G. Nabi, “Multimodal EEG-Based Classification of Alzheimer’s and MCI Using Olfactory Event-Related Potentials and Transformers,” *Brain-Apparatus Communication*, 2025.
3. I. Ashfaq, H. Saeed, Z. Uddin, A. Abid and **N. Fatima**, “Machine Learning-Based Classification for High-Efficiency Perovskite Solar Cell Material Screening,” in *Proc. 2026 Int. Conf. on Robotics and Automation in Industry (ICRAI)*, pp. 1–8, IEEE, May 2026.

HONORS & AWARDS

- **Excellence in Neuroscience Research**, KICS-UET Lahore June 2025  
*Recognized for contributions to AI/ML pipelines, signal processing, and innovative computational methods.*
- **Chief Minister Punjab’s Honhaar Scholarship**, Government of Punjab May 2025  
*Awarded for achieving one of the highest CGPAs (top 1%) in Computer Engineering at UET Lahore.*
- **Top 6 at Optimized AI Conference 2025**, Traversaal.ai Mar 2025  
*Team TROJAN\_AI ranked among Top 6 out of 200+ global teams for optimization and AI innovation.*
- **CS50x Puzzle Day 2025**, Harvard & MIT (Cambridge) Apr 2025  
*Recognized for outstanding performance in problem-solving, teamwork, and analytical thinking.*
- **National AI Wrapper Competition 2025**, Pakistan 2025  
*The IndusTwin project was selected among the top entries from 2,889 AI project ideas submitted by participants across 105+ cities and over 1,700 applicants nationwide, recognizing its potential for AI-driven climate resilience and flood prediction in Pakistan.*

## RELEVANT CERTIFICATES

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|                                                            |                             |
|------------------------------------------------------------|-----------------------------|
| Supervised Machine Learning: Regression and Classification | by DeepLearning.AI          |
| Deep Learning with PyTorch: Image Segmentation             | by Coursera Project Network |
| Introduction to Artificial Intelligence                    | by Google                   |
| Fundamentals of Machine Learning                           | by Microsoft                |
| The Nuts and Bolts of Machine Learning                     | by Google                   |
| Crash Course on Python                                     | by Google                   |
| SQL (Basic, Intermediate and Advanced)                     | by HackerRank               |
| AI Agents Fundamentals                                     | by Hugging Face             |
| Intro to Deep Learning                                     | by Kaggle                   |

## INVITED TALKS

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- **AI for Brain: Unlocking Cognitive Insights with Machine Learning** | Slides  
Invited Talk at *ACM UET Lahore Chapter* August 2025  
*Engaged 100+ students and faculty in ML applications for neuroscience and predictive modeling.*
- **Data-Centric AI: Why Better Data Beats Bigger Models** | Slides  
Invited Talk at *Google Developer Group (GDG), UET Lahore* August 2025  
*Addressed 50+ AI enthusiasts on data quality and intelligent system development.*

## VOLUNTEER EXPERIENCE

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**Committee Chair (Publicity) & Member** Oct 2025 – Present  
*Mediterranean Conference on Emerging Technologies and Systems (MCETS)*

- Serving as **Publicity Chair**, leading outreach and visibility strategies for an international academic conference.
- Coordinating with organizing committees and managing communication channels to enhance global participation.

**Member, Society of Artificial Intelligence** Aug 2025 – Present  
*International Association of Engineers (IAENG)*

- Actively engaged in **research dissemination**, conferences, and technical forums within the AI community.
- Contributing to discussions on **ethical and impactful AI** across academia and industry.

**Content Writer** May 2024 – Jun 2024  
*Dr Coders*

- Created engaging educational content on **programming and coding concepts** for aspiring developers.
- Simplified complex topics into concise, accessible formats for broader audience understanding.
- Collaborated with a team to produce structured learning resources.

**Ambassador** Mar 2024 – Apr 2024  
*AIRCI Expo*

- Led initiatives to **connect industry professionals** and foster collaboration during the expo.
- Executed targeted outreach and content strategies to **increase engagement and event reach**.

**Student Volunteer (UTS 4.0)** Feb 2024  
*ACM UET Lahore*

- Contributed to the **event décor team**, executing creative concepts to enhance event experience.
- Collaborated effectively within a team to deliver a visually impactful environment.

## SUMMER SCHOOLS & ACADEMIC TRAINING

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- **Computational Neuroscience, Neurotechnology and Neuro-inspired AI (ISRC-CN3) Summer School 2026**  
*Intelligent Systems Research Centre, Ulster University, UK* August 2026  
*Selected as an online participant for an international summer school focused on computational neuroscience, neurotechnology, brain-inspired AI, and intelligent systems research.*

- **ISDSA 2026 Online Workshops**

*International Chinese Statistical Association (ICSA) and International Society for Data Science and Analytics (ISDSA)*  
*Jul 2026*

- EEG Data Analysis Using R
- Trustworthy AI: Foundations, Methods, and Emerging Challenges
- Introduction to Dynamic Network Analysis